(09/05) CHE 210 - HW1

(1) Murphy 1.2 Solving using matrices
Nitroglycerin [$C_3H_5(NO_3)_3$] is an explosive that decomposes to CO_2, H_2O, N_2, and O_2. Write the balanced chemical equation of nitroglycerin.

$\underline{r_1}\, C_3H_5(NO_3)_3 \rightarrow \underline{r_2}\, CO_2 + \underline{r_3}\, H_2O + \underline{r_4}\, N_2 + \underline{r_5}\, O_2$

C: $3r_1 \rightarrow r_2$ ✓ $r_1 = 1$ C: $r_2 = 3$

H: $5r_1 \rightarrow 2r_3$ ✓   H: $2r_3 = 5$

N: $3r_1 \rightarrow 2r_4$ ✓   N: $2r_4 = 3$

O: $9r_1 \rightarrow 2r_2 + r_3 + 2r_5$ ✓   O: $2r_2 + r_3 + 2r_5 = 9$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 2 & 1 & 0 & 2 \end{bmatrix}\begin{bmatrix} r_2 \\ r_3 \\ r_4 \\ r_5 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 3 \\ 9 \end{bmatrix} \qquad r = a^{-1}B \qquad \begin{bmatrix} r_2 \\ r_3 \\ r_4 \\ r_5 \end{bmatrix} = \begin{bmatrix} 3 \\ 5/2 \\ 3/2 \\ 1/4 \end{bmatrix}$$

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$C_3H_5(NO_3)_3 \rightarrow 3CO_2 + \frac{5}{2}H_2O + \frac{3}{2}N_2 + \frac{1}{4}O_2$

$\boxed{4C_3H_5(NO_3)_3 \rightarrow 12CO_2 + 10H_2O + 6N_2 + O_2}$ ✓

(2) Murphy 1.15. Excess nitrate in well water
$HNO_3 + CH_3OH \rightarrow C_3H_7NO_2 + CO_2 + H_2O$
*Eq. written not balanced. Find stoich. coeff. Calc. the grams of methanol required to reduce the nitrate content (as HNO_3) of 10 L of well water from 54 mg/L to 10 mg/L.

$\underline{r_1}\, HNO_3 + \underline{r_2}\, CH_3OH \rightarrow \underline{r_3}\, C_3H_7NO_2 + \underline{r_4}\, CO_2 + \underline{r_5}\, H_2O$

$r_1 = 1$

C: $r_2 \rightarrow 3r_3 + r_4$ ✓   C: $-r_2 + 3r_3 + r_4 = 0$

H: $r_1 + 4r_2 \rightarrow 7r_3 + 2r_5$ ✓   H: $-4r_2 + 7r_3 + 2r_5 = 1$

N: $r_1 \rightarrow r_3$ ✓   N: $r_3 = 1$

O: $3r_1 + r_2 \rightarrow 2r_3 + 2r_4 + r_5$ ✓   O: $-r_2 + 2r_3 + 2r_4 + r_5 = 3$

$$\begin{bmatrix} -1 & 3 & 1 & 0 \\ -4 & 7 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ -1 & 2 & 2 & 1 \end{bmatrix}\begin{bmatrix} r_2 \\ r_3 \\ r_4 \\ r_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 3 \end{bmatrix} \rightarrow \begin{bmatrix} r_2 \\ r_3 \\ r_4 \\ r_5 \end{bmatrix} = \begin{bmatrix} 10/3 \\ 1 \\ 1/3 \\ 11/3 \end{bmatrix}$$

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$HNO_3 + 10/3\, CH_3OH \rightarrow C_3H_7NO_2 + 1/3\, CO_2 + 11/3\, H_2O$

$3HNO_3 + 10CH_3OH \rightarrow 3C_3H_7NO_2 + CO_2 + 11H_2O$ ✓

$$\frac{0.044\,g\,HNO_3}{L} \times \frac{10\,L}{1} \times \frac{1\,mol\,HNO_3}{63.015\,g\,HNO_3} \times \frac{10/3\,mol\,CH_3OH}{1\,mol\,HNO_3} \times \frac{32\,g\,CH_3OH}{1\,mol\,CH_3OH}$$

$\boxed{= 0.746\,g\,CH_3OH}$ ✓

(3) Murphy 1.17 - Carbon nanotubes proposed as a novel way to store hydrogen safely/compactly in solid form at room temperature and pressure. Several researchers have reported synthesizing nanotubes containing 1% - 8% by weight of hydrogen. One group claimed that certain graphite nanofibers can store hydrogen at levels exceeding 50 wt %, but others were skeptical. *What is the wt% hydrogen ~~at levels~~ in common liquids and solids such as ethanol (C_2H_5OH), methane (CH_4), water (H_2O) and glucose ($C_6H_{12}O_6$)? Use result of your calc. to evaluate the importance/validity of the claims regarding the hydrogen storage potential of nanotubes.

Ethanol (C_2H_5OH) = 46.07 g/mol

$$\frac{6 \text{ mol H} \times 1 \text{ g/mol of H}}{1 \text{ mol } C_2H_5OH \times 46.1 \text{ g/mol of } C_2H_5OH} \times 100 \boxed{= 13.02\%}$$

Methane (CH_4) = 16.04 g/mol

$$\frac{4 \text{ mol H} \times 1 \text{ g/mol of H}}{1 \text{ mol } CH_4 \times 16.04 \text{ g/mol of } CH_4} \times 100 \boxed{= 24.94\%}$$

Water (H_2O) = 18.02 g/mol

$$\frac{2 \text{ mol H} \times 1 \text{ g/mol of H}}{1 \text{ mol } H_2O \times 18.02 \text{ g/mol of } H_2O} \times 100 \boxed{= 11.10\%}$$

Glucose ($C_6H_{12}O_6$) = 180.16 g/mol

$$\frac{12 \text{ mol H} \times 1 \text{ g/mol of H}}{1 \text{ mol } C_6H_{12}O_6 \times 180.16 \text{ g/mol of } C_6H_{12}O_6} \times 100 \boxed{= 6.66\%}$$

Because the molecular weight of Hydrogen is so small compared to Carbon and Oxygen, it is impossible to store Hydrogen at a level exceeding 50 wt.%. Thus, the statement above is not true.

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(5) A tank can be filled w/ liquid water from three diff. water supply lines labeled A, B, and C. The flow rates from each supply line is constant. The volume of the tank is unknown. When supply lines A and B are used simultaneously to fill the tank, the filling process takes half an hour. Similarly, when A and C are used, it takes three-quarters of an hour. And when B and C are used, it takes one hour.

(a) How long does it take to fill the tank when all three supply lines are used simultaneously?

$V = 1$

$QV = t$

$A + B \; @ \; t = 0.5\,hr \qquad (Q_A + Q_B)(0.5\,hr) = V$

$A + C \; @ \; t = 0.75\,hr \rightarrow (Q_A + Q_C)(0.75\,hr) = V$

$B + C \; @ \; t = 1.0\,hr \qquad (Q_B + Q_C)(1.0\,hr) = V$

$\frac{1}{2} Q_A + \frac{1}{2} Q_B = 1$

$\frac{3}{4} Q_A + \frac{3}{4} Q_C = 1$

$Q_B + Q_C = 1$

$Q_B = 1 - Q_C$

$\frac{1}{2} Q_A + \frac{1}{2}(1 - Q_C) = 1$

$\frac{1}{2} Q_A + \frac{1}{2} - \frac{1}{2} Q_C = 1$

$\frac{1}{2} Q_A + \frac{1}{2} = 1 + \frac{1}{2} Q_C$

$-\frac{1}{2} \qquad -\frac{1}{2}$

$\frac{1}{2} Q_A = \frac{1}{2} + \frac{1}{2} Q_C$

$Q_A = 1 + Q_C$

$\frac{3}{4}(1 + Q_C) + \frac{3}{4} Q_C = 1$

$\frac{3}{4} + \frac{3}{4} Q_C + \frac{3}{4} Q_C = 1$

$\frac{3}{2} Q_C = \frac{1}{4} \rightarrow Q_C = \frac{1}{6}$

$Q_B + Q_C = 1 \rightarrow Q_B = 1 - \frac{1}{6} = \frac{5}{6}$

$\frac{1}{2} Q_A + \frac{1}{2} Q_B = 1$

$\frac{1}{2} Q_A + \frac{1}{2}(\frac{5}{6}) = 1 \rightarrow Q_A = \frac{7}{6}$

$\left[Q_A = \frac{7}{6},\; Q_B = \frac{5}{6},\; Q_C = \frac{1}{6} \right]$ [tank.V/hr.]

$(Q_A + Q_B + Q_C)t = V \rightarrow \left[\frac{7}{6} + \frac{5}{6} + \frac{1}{6}\right] t = 1\,\text{tank}\cdot V$

$t = 6/13\,hr$

$\boxed{t \approx 0.462\,hr.}$

(b) How long does it take for each supply line alone to fill the tank?

Line A: $Q_A t_A = V \rightarrow \frac{7}{6} \frac{\text{tank}\cdot V}{hr} = 1\,\text{tank}\cdot V$

$\boxed{t_A = 0.86\,hr.}$

Line B: $\frac{5}{6} \frac{\text{tank}\cdot V}{hr.} = 1\,\text{tank}\cdot V \rightarrow \boxed{t_B = 1.2\,hr.}$

Line C: $\frac{1}{6} \frac{\text{tank}\cdot V}{hr} = 1\,\text{tank}\cdot V \rightarrow \boxed{t_C = 6\,hr.}$

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